

ANEMIA

- **Definition:** Anemia is a blood disorder, characterised by reduction in:
 - a. Red blood cell (RBC) count
 - b. Hemoglobin (Hb) content
 - c. Packed cell volume (PCV)

Because of :—

- Decreased production of RBC
- Increased destruction of RBC
- Excess loss of blood from body

CLASSIFICATION OF ANEMIA

- Classified by 2 methods:
 - a. Morphological classification
 - b. Etiological classification

1. MORPHOLOGICAL CLASSIFICATION

- Depends upon:
 - **Size of RBC** → Determined by MCV (*Mean Corpuscular Volume*)
 - **Color of RBC** → Determined by MCHC (*Mean Corpuscular Hb Concentration*)
- **Types :—**
 - a. **Normocytic Normochromic Anemia:** Size & color are normal; number of RBCs is less.
 - b. **Macrocytic Normochromic Anemia:** Size is large; color is normal; RBC count is less.
 - c. **Macrocytic Hypochromic Anemia:** Size is large; MCHC is low (pale colour).
 - d. **Microcytic Hypochromic Anemia:** Small size + less color.

2. ETIOLOGICAL CLASSIFICATION

(Study of cause or origin)

- a. Hemorrhagic anemia
- b. Hemolytic anemia
- c. Nutrition deficiency anemia
- d. Aplastic anemia
- e. Anemia of chronic disease

A. Hemorrhagic Anemia

- Anemia due to **excessive loss of blood**.
- **Types :—**
 - **Acute Hemorrhage:**
 - Loss of large quantity of blood as in case of accident.
 - Decreased RBC count causes hypoxia → corrected within 4 – 6 weeks.
 - **RBCs are:** Normocytic, normochromic.
 - **Chronic Hemorrhage:**
 - Loss of blood by internal / external bleeding over a long period of time (*e.g., peptic ulcer, menorrhagia*).
 - **RBCs are:** Microcytic, hypochromic.
 - Replacement of RBCs does not occur quickly.

B. Hemolytic Anemia

- Anemia due to **excessive destruction of RBCs** (*not compensated by increased RBC production*).
- **Types :—**
 - **Extrinsic Hemolytic Anemia:**
 - Caused by destruction of RBCs by external factors (*e.g., hypersplenism, liver failure, renal disorder*).
 - **Intrinsic Hemolytic Anemia:**
 - Caused by destruction of RBCs becoz of defective RBCs (*e.g., sickle cell anemia, thalassemia*).

C. Nutrition Deficiency Anemia

- Anemia occurs due to deficiency of a nutritive substance necessary for erythropoiesis.
- **Types :—**
 - **Iron (Fe) Deficiency Anemia:** RBCs are microcytic & hypochromic.
 - **Vitamin B12 Deficiency (Pernicious Anemia / Addison's Anemia):** RBCs are macrocytic & normochromic / hypochromic.
 - **Folic Acid Deficiency (Megaloblastic Anemia):** RBCs are megaloblastic & hypochromic.

D. Aplastic Anemia

- Anemia due to disorder of red bone marrow.
- Red bone marrow reduced & replaced by fatty tissues.
- **Causes:** Bone marrow disorder occurs in repeated exposure to X-ray, tuberculosis, viral infections (*hepatitis*).
- **RBCs are:** Normocytic & normochromic.

E. Anemia of Chronic Diseases

- Anemia develops after few months of sustained disease.
- **RBCs are:** Normocytic & normochromic.
- **Occurs in chronic infections :—**
 - Tuberculosis
 - Chronic renal failure
 - Cancer disorders
 - Rheumatoid arthritis

SIGNS AND SYMPTOMS OF ANEMIA

- a. **Skin and Mucous Membrane** → Color of skin and mucous membrane becomes pale.
- b. **Cardiovascular System** → Increase in HR & cardiac output.
- c. **Respiration** → Increase in rate & force of respiration (*sometimes leads to breathlessness, dyspnea*).
- d. **Digestion** → Anorexia, nausea, vomiting, abdominal discomfort.

- e. **Kidney** → Renal function disturbed.
- f. **Reproductive System** → In female, menstrual cycle disturbed.
- g. **Neuromuscular System** → Headache, lack of concentration, restlessness, irritability, vertigo.
- h. **Muscles** → Muscles become weak & patient feels lack of energy and fatigued quite often & quite easily.